Wind Load vs. Wind Speed

Wind load on surface - Wind load calculator.

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When moving air - wind - is stopped by a surface - the dynamic energy in the wind is transformed to pressure. The pressure acting the surface transforms to a force

$$F_w = p_d A$$

$$= 1/2 \rho v^2 A \qquad (1)$$

where

F_w = wind force (N)

A = surface area (m^2)

p_d = dynamic pressure (Pa)

ρ = density of air (kg/m^3)

v = wind speed (m/s)

Note - in practice wind force acting on a object creates more complex forces due to drag and other effects.

Wind Load Calculator

1.2 *air density (kg/m^3)*

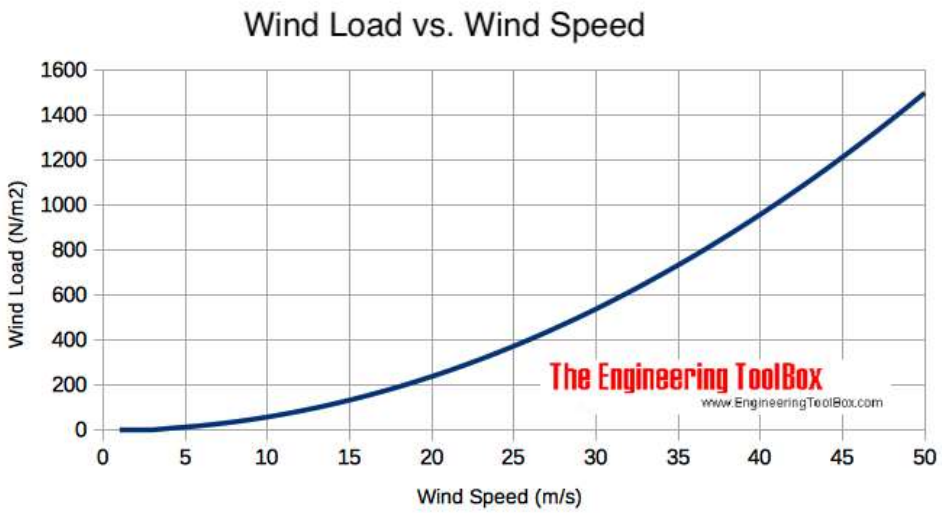
40 *wind speed (m/s)*

10 *area (m^2)*

Calculate!

- Dynamic pressure (N/m^2, Pa): **960**
- Wind Load (N): **9600**

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Wind Speed *(m/s)* (ft/min)	**Wind Load**[1)] *(Pa)* (psi)
1	0.6
2	2.4
3	5.4
4	9.6
5	15
6	22
7	29
8	38
9	49
10	60
11	73
12	86
13	101
14	118
15	135
16	154
17	173
18	194
19	217
20	240
21	265
22	290
23	317
24	346
25	375
26	406
27	437
28	470
29	505
30	540
31	577
32	614
33	653
34	694
35	735
36	778
37	821
38	866

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Wind Speed (m/s) (ft/min)	Wind Load[1)] (Pa) (psi)
45	1215
46	1270
47	1325
48	1382
49	1441
50	1500

1) density of air 1.2 kg/m^3

- 1 m/s = 3.6 km/h = 196.85 ft/min = 2.237 mph
- 1 Pa = 1 N/m^2 = 1.4504x10^{-4} psi (lb/in^2)

Example - Hurricane Wind Load acting on a Wall Surface

A hurricane with wind speed 35 m/s is acting on a 10 m^2 wall. The dynamic force can be calculated as

$$F_w = 1/2\ \rho\ v^2\ A$$

$$= 1/2\ (1.2\ kg/m^3)\ (35\ m/s)^2\ (10\ m^2)$$

$$= \underline{7350}\ N$$

$$= \underline{7.35}\ kN$$

Or - from the table above the wind load per square metre is 735 N/m^2. The total load on the wall can be calculated as

$$(735\ N/m^2)\ (10\ m^2) = \underline{7350}\ N$$

A hurricane acting on a 10 m^2 wall creates a force equal to the weight of aprox. 750 kg.

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Related Topics

- **Mechanics** - Forces, acceleration, displacement, vectors, motion, momentum, energy of objects and more.

Related Documents

- **Air Curtains or Air Screens** - Calculate air curtains or air screens in open doorways used to keep acceptable indoor comfort in buildings.
- **Beaufort Wind Scale** - The Beaufort description and observation scale for wind and wind speed.
- **Distance Traveled vs. Speed and Time - Calculator and Chart** - Velocity plotted in time used diagram.
- **Drag Coefficient** - The drag coefficient quantifies the drag or resistance of an object in a fluid environment.
- **Dynamic Pressure** - Dynamic pressure is the kinetic energy per unit volume of a fluid in movement.

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- **Wind Energy** - Available wind energy.
- **Wind Power** - Power generation from wind.
- **Wind Shear** - Wind slowed down at surface by trees and obstructions.
- **Wind Turbine Power Calculator** - Online wind turbine calculator.

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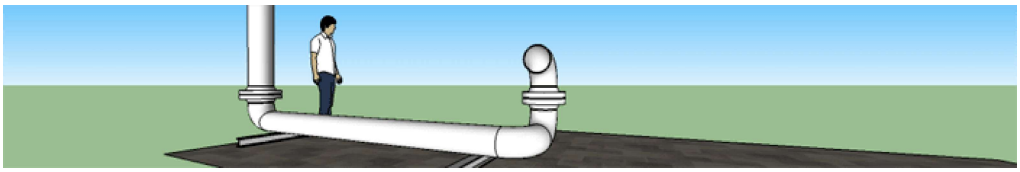
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- **Thermodynamics**
- **Water Systems**

Unit Converter

Temperature

0.0

- ○ ^oC
- ○ ^oF

Convert!

Length

1.0

- ○ *m*
- ○ *km*
- ○ *in*
- ○ *ft*
- ○ *yards*
- ○ *miles*
- ○ *naut miles*

Convert!

Area

1.0

- ○ m^2
- ○ km^2
- ○ in^2
- ○ ft^2
- ○ $miles^2$
- ○ *acres*

Convert!

Volume

1.0

- ○ m^3
- ○ *liters*
- ○ in^3
- ○ ft^3
- ○ *us gal*

Convert!

Weight

1.0

- ○ kg_f
- ○ *N*
- ○ lb_f

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- (•) *m/s*
- ( ) *km/h*
- ( ) *ft/min*
- ( ) *ft/s*
- ( ) *mph*
- ( ) *knots*

Convert!

Pressure

1.0

- (•) *Pa (N/m^2)*
- ( ) *bar*
- ( ) *mm H_2O*
- ( ) *kg/cm^2*
- ( ) *psi*
- ( ) *inches H_2O*

Convert!

Flow

1.0

- (•) *m^3/s*
- ( ) *m^3/h*
- ( ) *US gpm*
- ( ) *cfm*

Convert!

Scientific Online Calculator

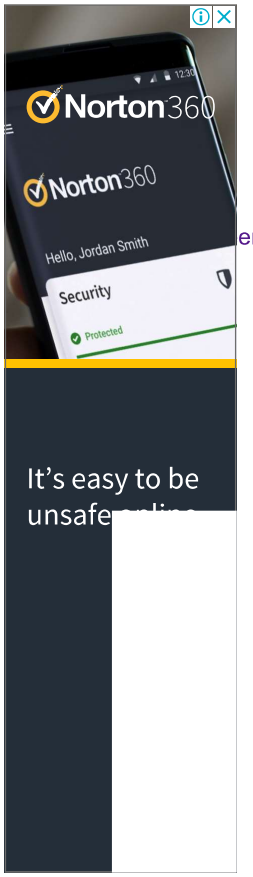


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